AN INTEGRATED APPROACH FOR IMPACT ASSESSMENT OF WATER HARVESTING TECHNIQUES IN DRY AREAS: THE CASE OF OUED OUM ZESSAR WATERSHED (TUNISIA)

M. OUESSAR[1,*], M. SGHAIER[1], N. MAHDHI[1], F. ABDELLI[1], J. DE GRAAFF[2], H. CHAIEB[3], H. YAHYAOUI[3] and D. GABRIELS[4]

[1]*Institut des Régions Arides (IRA), Médenine, Tunisia;* [2]*Erosion and SWC Group, Wageningen University, Wageningen, The Netherlands;* [3]*Direction Générale des Ressources en Eau, Ministère de l'Agriculture, Tunis, Tunisia;* [4]*Department of Soil Management and Soil Care, Ghent University, Gent, Belgium*

*(*author for correspondence, e-mail: Ouessar.Mohamed@ira.rnrt.tn)*

Abstract. In the arid regions of Tunisia, considerable investments are being made to maintain the old water harvesting techniques and introduce new ones to capture the scarce amount of rainwater (100 mm to 230 mm annually) for agricultural and domestic purposes. However, no detailed assessment of the multiple effects and the costs and benefits of these techniques have been made so far. This paper summarizes the results of an in depth investigation of the multiple impacts (runoff mobilization, ground water recharge, agro-socio-economic impacts) of the water harvesting works undertaken in the watershed of oued Oum Zessar (southeastern Tunisia). The importance of interdisciplinary and integrated approaches was revealed through this detailed impact assessment and economic evaluation. In fact, the profitability of the water harvesting works depends largely on the criteria chosen. However, further refinements are needed to better include all possible impacts (positive and negative) that occur as a result of the installation of the water harvesting structures.

Keywords: arid lands, economics, impact assessment, Tunisia, water harvesting

1. Introduction

Water is vital to every human community and is an essential resource for economic development, agricultural productivity, industrial growth, and human well-being. The availability of a clean, safe, and secure water source has been, and will always be, a major concern for human populations.

Because of its location north of the 30th parallel, the climate of Tunisia is influenced by the Mediterranean Sea and the caprices of the Sahara Desert. The arid, semi-arid and desert bioclimates cover more than two-thirds of its area (Floret and Pontanier, 1982). The rainfall regime is characterized by its scarcity, variability, torrential nature and poor distribution. In the dry regions of the country the real evapotranspiration/potential evapotranspiration (ETR/ETP) is generally very low and does not exceed 30% which reflects a high deficit in the water balance (Hénia, 1993).

Tunisia occupies a strategic area between the eastern and the western basin of the Mediterranean Sea. Numerous populations (Berbers, Numids, Phoenicians,

Environmental Monitoring and Assessment **99:** 127–140, 2004.
© 2004 *Kluwer Academic Publishers. Printed in the Netherlands.*

Romans, Arabs) have lived in this country for thousands of years. The environmental challenges (climate, geomorphology, geology) did not prevent them from building flourishing civilizations. Their heritage and many still functioning structures are good indicators of their excellent adaptation to the natural environment (Ennabli, 1993; Ben Mechlia and Ouessar, 2002).

Rainfed farming in Tunisia, which represents an important component of the agricultural production system, has been supported mainly by the various water harvesting techniques (WHT) developed since the pre-Roman times, as it is the case in many other countries (Harris, 1991). A wide variety of small to medium-sized hydraulic structures were introduced throughout the centuries to make the land productive, irrespective of its geographical location (El Amami, 1984).

In the southeastern region of the country where the annual rainfall ranges between 100 mm and 230 mm, the areas treated with traditional WHT (mainly *jessour*: small retention dams) were limited to the upper parts of the *wadis* catchment zones. However, with independence there has been a gradual extension of the cultivated fields, mainly olive tree groves, to neighboring areas, thanks to the construction of *tabias* (the same structure as *jessour* but situated in foothills) and water spreading structures in the foothills and surrounding plains *(jeffara*), normally exploited as rangelands. In parallel, especially during the last decade, the soil and water conservation service of the Ministry of Agriculture has also introduced new techniques (*gabion* units, ground water recharge wells, etc.) as part of the implementation of the National Soil and Water Conservation and the Water Resources Mobilization Strategies (Mini. Agri., 1990 a,b).

As far as watersheds are concerned, the enhancement of traditional techniques has induced multiple physical and socio-economic impacts that need an overall assessment (Fauck *et al*., 1991). This detailed impact assessment and economic evaluation aims to unravel why certain traditional measures have been successful in the past, and how effective and efficient new techniques are. In addition, it addresses the question of, to what extent the soil and water conservation programs could contribute, or not, to the improvement of the wellbeing of local populations. A tool for decision-makers can then be developed to assess under which agro-ecological and socio-economic conditions investment in water harvesting measures could be a viable undertaking in dry Mediterranean areas.

2. Materials and Methods

2.1. Study site

The Oued Oum Zessar watershed (SWS) was chosen as the case study area (Figure 1). Comprising 336 km^2, it is located in southeastern Tunisia and has an arid

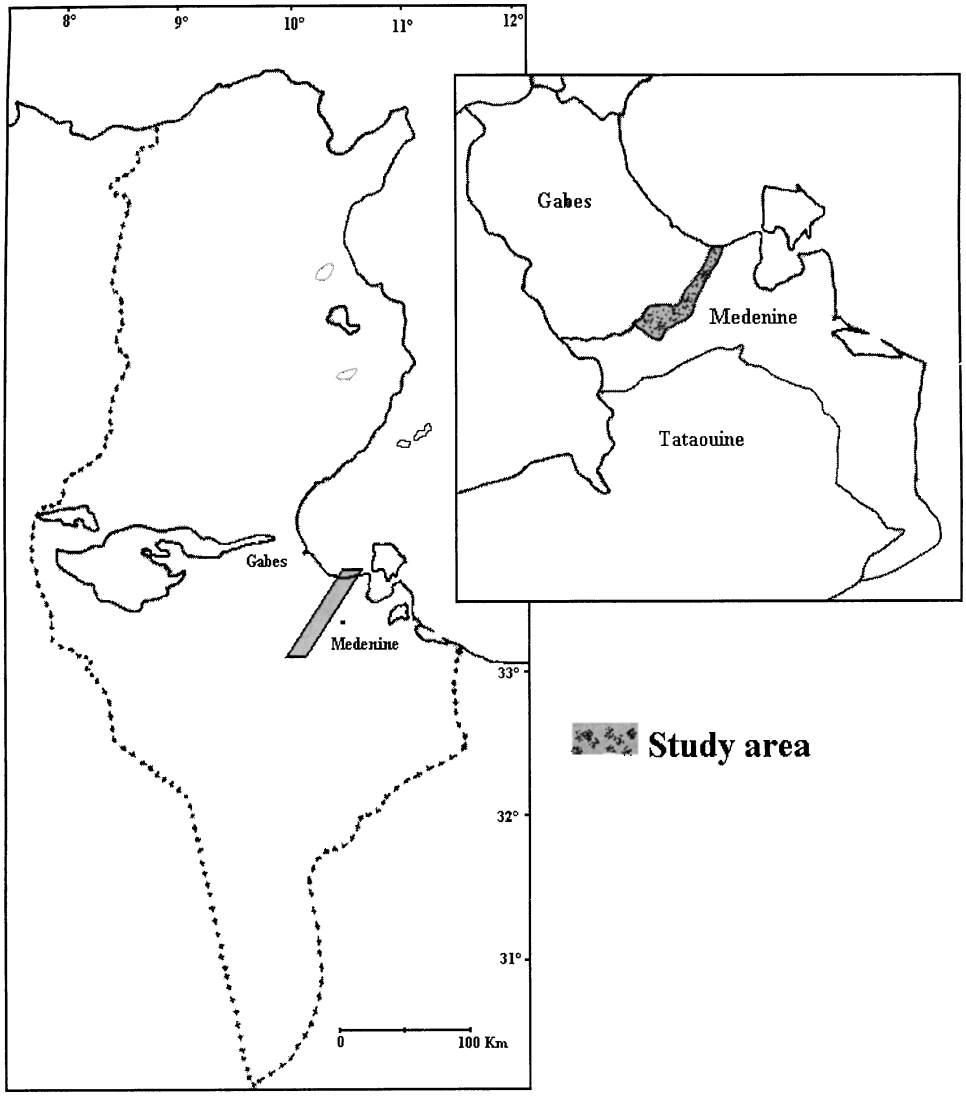

Figure 1. Study site.

Mediterranean bioclimate with an annual rainfall range of between 150 mm and 230 mm. It starts at the Matmata mountain range (Kef Ennsoura) and ends in the Mediterranean Sea. The oued Oum Zessar has three main tributaries: oued Negueb, oued Mogar and oued Hallouf. The hydraulic history of this SWS is very ancient and rich, as witnessed by the remains of a dam supposedly built in the Roman era near the village of Koutine, and the ancient terraces encountered in the hills of oued Nagab. At present, the upstream area is terraced with *jessour* while the intermediate zones incorporate *tabias* and floodwater diversion structures. On the

TABLE I
General characteristics of the study watershed

Characteristics	
Area (km^2)	367
Annual rainfall (mm)	180
Mean annual temperature (°C)	20
Altitude (m)	0–690
Population (inhabitants)	24 188

main *wadi* bed, several gabion units were built for the recharge of the groundwater aquifer of Zeuss–Koutine (Table I).

2.2. Approach and methodology

The overall research methodology is based on an interdisciplinary and integrated approach divided into two main interactive phases as described in Figure 2.

2.2.1. *Inventory*

An inventory of WHT was compiled through literature reviews and field investigations in addition to a review of available archives at the regional department of agriculture.

2.2.2. *Soil Erosion and Water Balance*

An event-based physical soil transport model (STM) for determining the overland flow processes and the resulting soil transport, net erosion and deposition developed by Biesemans (2000) was used for calculating the hydrological and erosion effects of each specific water harvesting technique. The model contains the possibility of using pedo-transfer functions if specific physical soil data are not available. The application was made on a *jessour* system in order to determine the amount of harvested water during a rainstorm as well as the soil erosion/deposition. Some field experiments (rainfall and infiltration simulations) were also performed. The soil water content was monitored using time domain reflectometry (TDR) tubes installed on the terraces of the structures.

2.2.3. *Ground Water Monitoring*

Since 1965, the ground water table of Zeuss-Koutine and its communicating aquifers (grès de trias) have been monitored monthly for water level and salinity fluctuations by a battery of 12 piezometers. The MULTIC (Derouiche, 1997) model was applied for recharge impact evaluation.

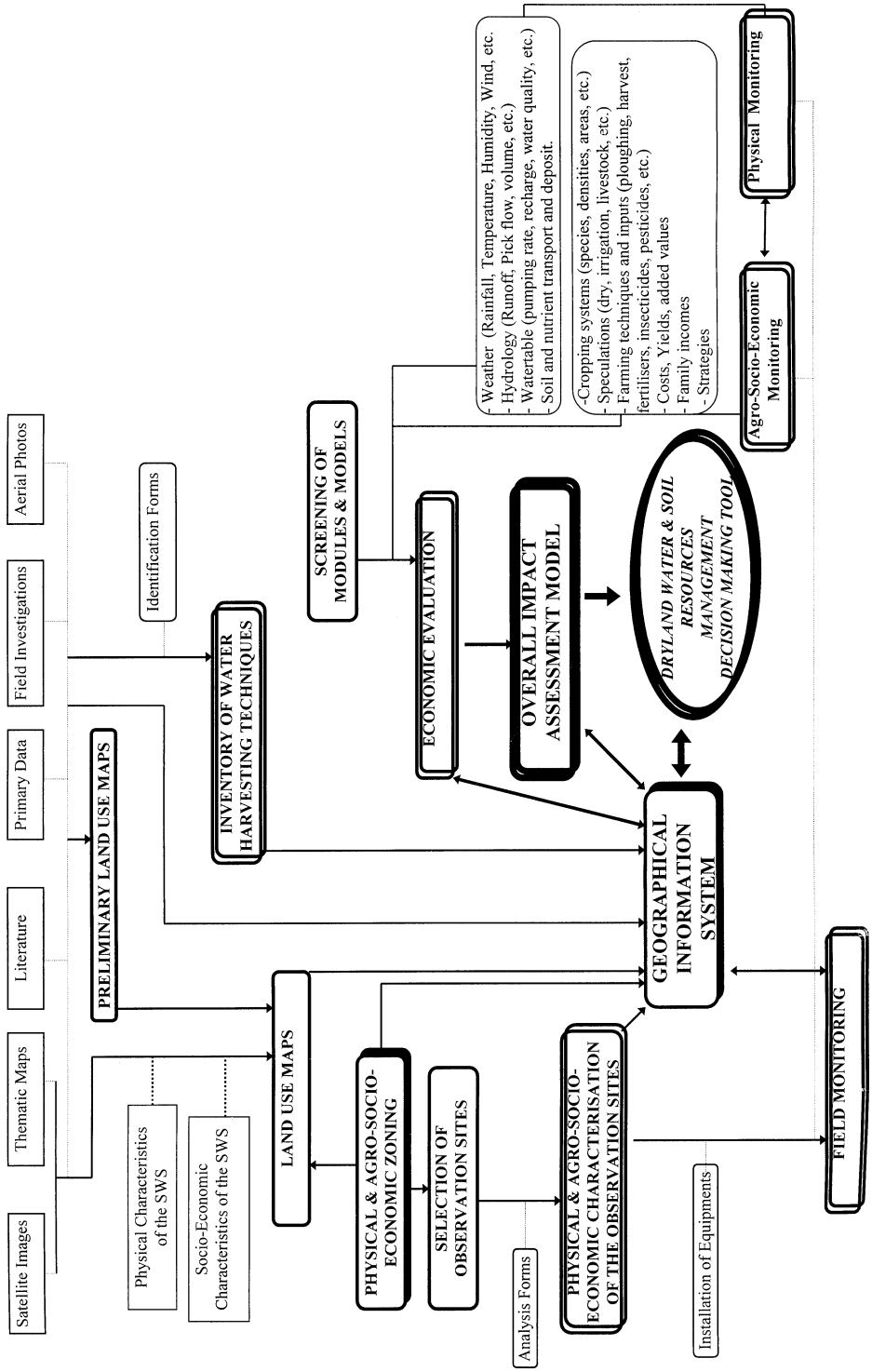

Figure 2. Methodology flow-chart (After WAHIA (1999)).

TABLE II
Interviewees by area

Interviewees	Up-stream	Down-stream
Beneficiaries of WHT	15	10
Users of surface wells	13	6
Irrigation farmers using drilled wells	–	5
Herdsmen	9	5
Fishermen	–	5

2.2.4. *Agro-Socio-Economic Assessment*
In order to have a general idea about the impact of WH works, agro-socio-economic surveys were carried out, both with beneficiaries and with potentially affected people. For this purpose, the SWS was divided into two main zones: an upstream area where the majority of WH measures are undertaken, and a downstream area, where off-site effects would likely be felt most. Amongst the people interviewed, the following groups were distinguished: beneficiaries of the WH measures, herdsmen, irrigation farmers depending on surface wells, irrigation farmers depending on drilled wells and fishermen (Table II).

Three evaluation methods were selected: Cost-Benefit Analysis (CBA), Multi-Criteria Analysis (MCA) and the model FORCES-MOD, which is in fact a form of Extended CBA, specifically applicable to forestry and soil conservation projects and programs. The applicability of each method was determined by the complexity it could handle and its data requirements.

- CBA was applied on the scale of individual WHT: *jessour* and tabia.
- MCA was applied on a watershed scale, taking into consideration different implementation intensities and a combination of different measures.
- FORCES–MOD was used to perform a scenario analysis at the level of the whole watershed.

Off-site effects included in the impact assessment were flood control (damage, inundation), increase in productive land acreage, lessening of erosion, ground water recharge and improvement of life quality of the inhabitants of the watershed.

The groups considered in the social CBA within FORCES-MOD and in MCA were:

1. beneficiaries of jessour,
2. beneficiaries of tabias,
3. irrigation farmers,
4. herdsmen,
5. fishermen.

TABLE III
Main data collected for impact assessment

Techniques	Jesser	Tabia	Gabion
Dimensions	X	X	X
Physical description	X	X	X
Land use	X	X	X
Effects			
Water balance	X	X	X
Sediment	X	X	X
Water quality	X	X	X
Costs/benefits			
Production systems	X	X	(X)
Input and output	X	X	–
Cost of techniques	X	X	X

Source. WAHIA (2000).

The FORCES-MOD model (FAO and World Bank, 1994) was employed to evaluate the construction of WH measures in the whole catchment area of oued Oum Zessar. This was done on five different levels:

1. financial analysis (FA),
2. conventional economic analysis (CEA),
3. extended economic analysis I (including tangible external effects) (EEA I),
4. extended economic analysis II (including tangible and intangible external effects) (EEA II),
5. socio-economic analysis (SEA).

2.3. Data collection

For an impact assessment of WHT, detailed physical and socio-economic data sets were required. The selected water harvesting techniques were: jessour, tabias and gabion structures. The data requirements for this impact assessment are summarized in Table III.

3. Results and Discussions

3.1. Inventory of water harvesting techniques

A wide variety of water harvesting techniques is found in the SWS and the study examined the effects of six of them.

3.1.1. *Terraces*

It seems that this technique is the oldest WHT adopted in the SWS and in the region. As in other regions of Tunisia (Raf Raf) and elsewhere (Yemen), they are encountered on steep slopes and are formed from small retaining walls made of rocks. As mentioned above, the remains of ancient terraces are found locally in the upper area of oued Nagab.

3.1.2. *Jessour*

Jessour are mainly found in the mountainous areas of Béni Khdache and Moggar This system is also an ancient WHT widespread in the region of the mountains of Matmata and is used in the inter mountain and hill water courses to intercept runoff and sediments. Jessour is the plural of a *jessour* which is a hydraulic unit made of three main components: a dike, a terrace and an impluvium.

- The dike *(tabia, sed, katra)* is a barrier created to block sediments and run-off water. Its body is made of earth equipped with a central *(masraf)* and/or lateral *(manfes)* spillway assuring the evacuation of excess water.
- The terrace is the area used for crops. It is formed progressively by the deposit of sediments. Generally, fruit trees (olive, fig, almond, date palm) and legumes (pea, chickpea, lentil, broad beans) are planted in the neighborhood of the dike while the remaining areas are cultivated with cereals (barley, wheat).
- The impluvium is the area destined for collecting and channelling flood water. It is contained by the natural watercourses.

3.1.3. *Tabias*

Tabias are essentially situated in areas with deep soils with a slope not exceeding 3%. In the Oum Zessar watershed this occurs in the area between Bhayra and Koutine. The tabia is formed by a principal bank of 50–150 m situated along contour lines with lateral bunds of about 30 m long at the ends. Water is stored until it reaches a height of 20 cm to 30 cm, after which it is diverted, either by a spillway or at the upper ends of the lateral bunds. The tabia gains its water directly from its impluvium (K values between 6 and 20) or by diversion of *wadi* runoff by a mgoud (traditional water spreading and diversion dike) (Alaya *et al.*, 1993). Annuals and fruit trees are generally cultivated on it.

3.1.4. *Cisterns*

These techniques, locally known as fesquia or majel, are built for the collection and storage of rainfall for watering animals, domestic use and irrigation. A cistern is a hole dug in the soil with a gypsic or cement coating to avoid vertical and lateral infiltration. Generally, each unit is made of three main components; the impluvium, the decantation basin, and the storage and pumping reservoir. Cisterns ranging from 5 to 50 m^3 can be found throughout the entire SWS.

3.1.5. *Gabion Units*
These structures are made of blocks of galvanized nets (gabions) filled with rocks. They are constructed in the oued beds. In general they have the form of a rectangular spillway and are used for two purposes:

- to slow down the runoff flow and thus to increase the infiltration rate to the underground water tables and/or
- to divert a portion of the runoff to neighboring cultivated fields (tabias).

These units are encountered on the main tributaries of oued Oum Zessar.

3.1.6. *Recharge Wells*
When the permeability of the underlying bedrock is judged to be too low, casting tubes are drilled into the rock to enhance the infiltration of runoff water to the water table. In our SWS, these recharge wells were installed behind gabion units at the level of Koutine.

3.1.7. *Developments*
During the last decade (1990–2000) the regional services of soil and water conservation executed two main national programs, namely the soil and water conservation strategy and the water resources development strategy. The works consisted mainly of the enhancement of jessour, tabias and terracing. The temporal distribution of the development is shown in Figure 3.

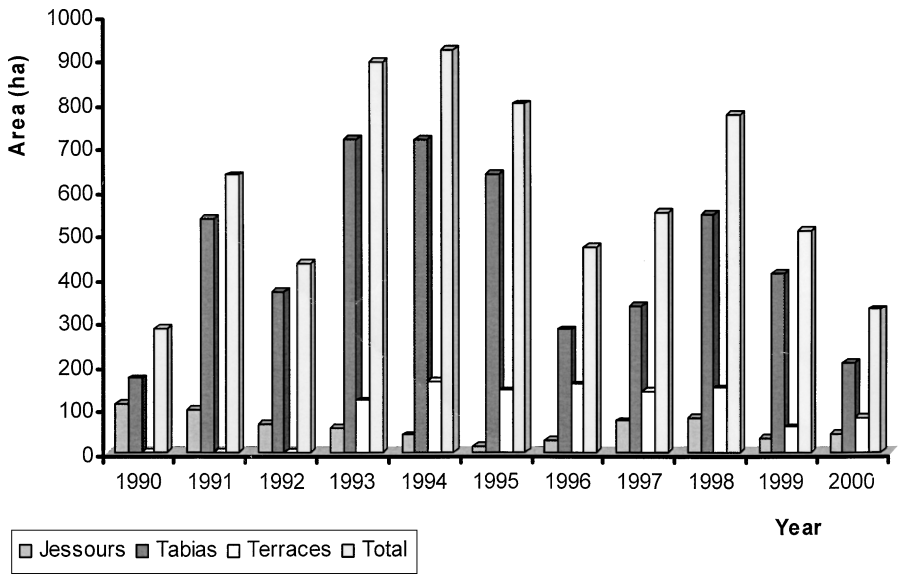

Figure 3. Works undertaken during the period 1990–2000.

Most of the work related to surface water was undertaken between 1990 and 1993. By the end of 2000, 243 recharge units and 27 spreading units had been renovated while three recharge wells had also been installed.

3.2. Hydrological and erosion effects

The WH structures are meant to collect runoff water coming from the surrounding catchment and also to trap eroded materials. An example of the results following a heavy downpour of 77 mm in an hour is given in Figure 4. The estimated harvested water was 2 340 mm with a deposition of 35 mm of soil.

The 'artificial' soil formed, plays an important role in water and nutrient supply of trees and crops as shown in Figure 5. The water content over a period of 3 yr remained above the PWP despite low rainfall figures.

3.3. Ground water replenishment

Looking at the results of the recharge influx for 3 yr (Figure 6), we can notice a gradual increase in the infiltration volume to the aquifer as a function of the progress of the water harvesting works implementation. In fact:

- In 1975, the watershed was partially renovated and the recharge was around 285L/s.
- In 1996, which coincides with the implementation of the soil and water conservation and water development strategies, the recharge was estimated at 301L/s.
- 2000 represents the year of achieving these strategies and the recharge reached 447L/s.

As far as the water salinity is concerned, the effects of the WT were not straightforward because of the geometry of the reservoir. Between 1982 and 2000, water quality has revealed a gradual increase in salinity (42 mg/l/yr) which reflects a vertical salinity exchange phenomenon. In fact, pumping gradually exhausts the fresh water which has infiltrated from surface water. This abstraction generates a depression cone and an upward movement of deep salty waters, inducing a gradual increase in the flow from the geological reserves of the water table.

3.4. Economic evaluation

The summarized results are shown in Table IV.

The enlarged CBA took into account environmental phenomena such as structural damages, flooding and aquifer recharge during the economic assessment. Financial criteria – Internal Rate of Return (IRR) and Net Present Value (NPV) progressed from low values in the financial analysis to higher levels when enlarging the analysis to include environmental and social aspects.

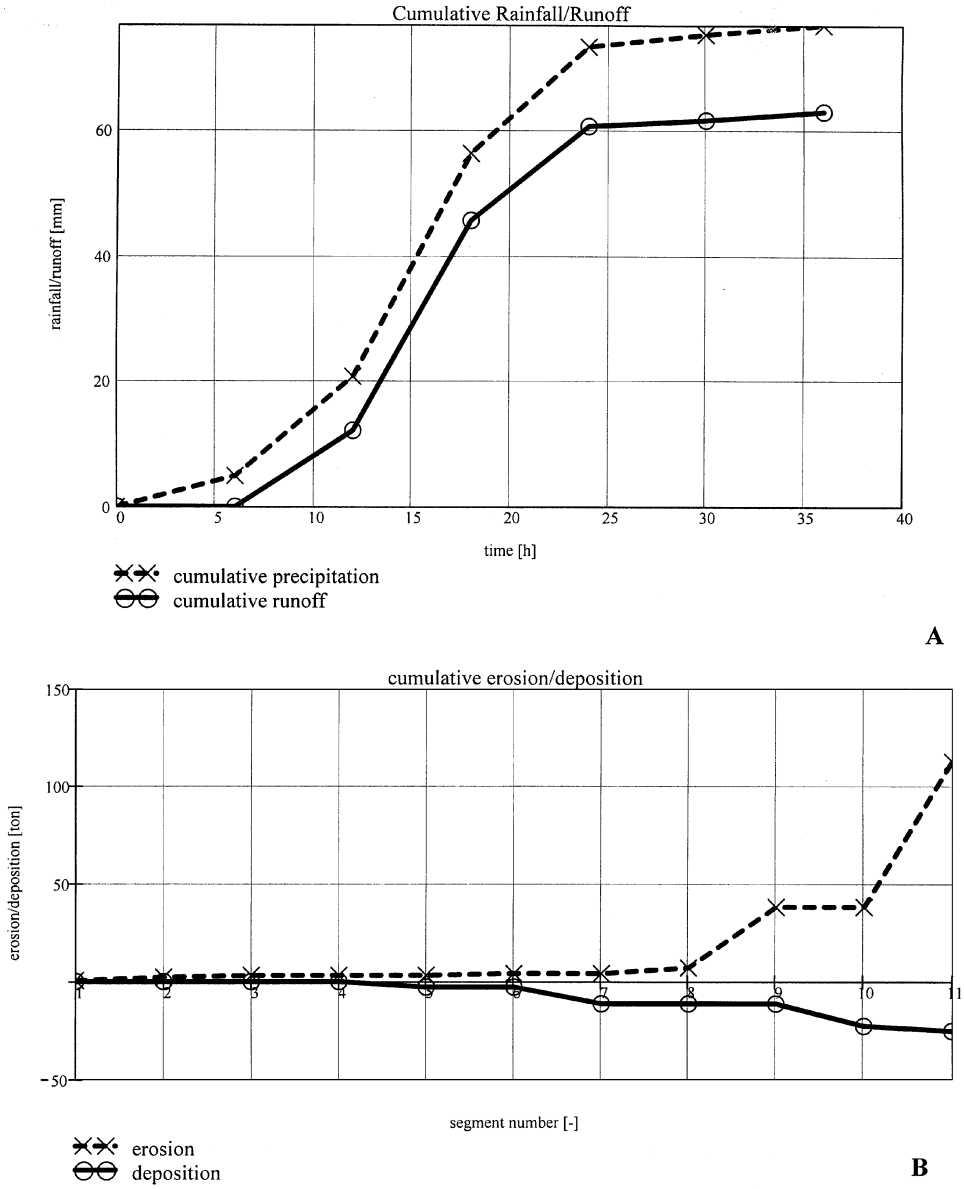

Figure 4. Cumulative precipitation/runoff (A) and erosion/deposition (B).

The economic analysis shows that the WHT works are profitable from a national economic point of view, with an IRR of 13.3%. The inclusion in the analysis of the off-site or downstream benefits of reducing flood damage to structures (EEA I) improves the economic profitability considerably, with an NPV (at 10.8% discount rate) of 0.79 million Tunisian dinars (MDT) and an IRR of 18.4%. And the

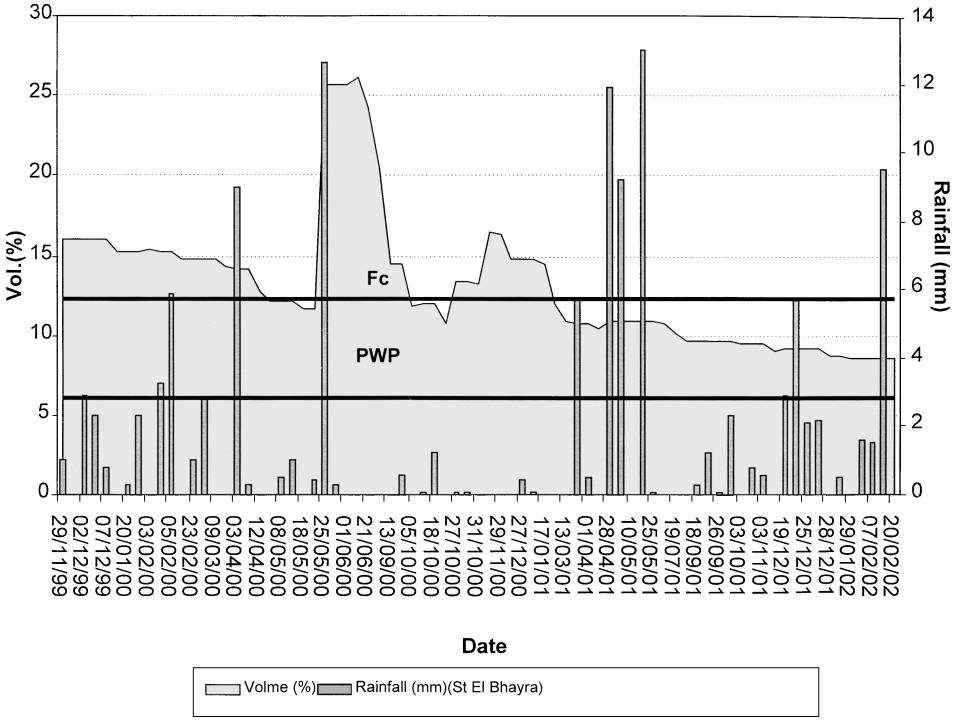

Figure 5. Soil water content of a terrace behind a gabion unit.

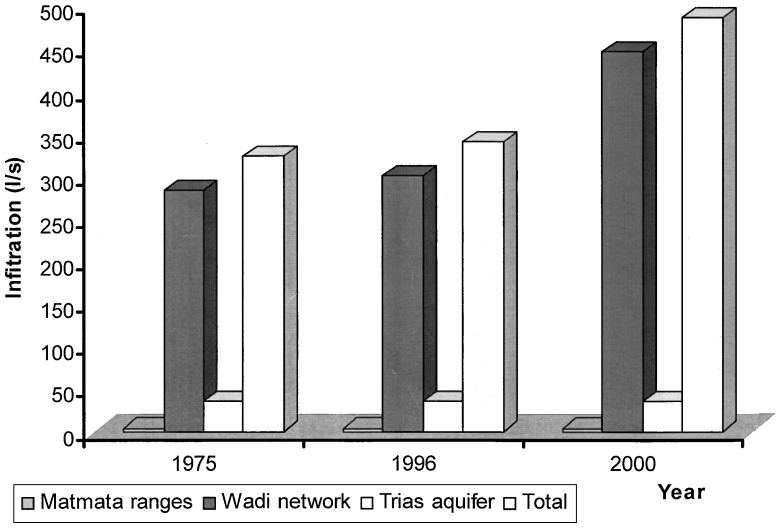

Figure 6. Ground water replenishment.

TABLE IV
FORCES-MOD application results

	NPV			IRR (%)
Discounting rate	10.8	8	13.8	
Financial analysis (FA)	−1.06	−0.70	−1.19	5.5
Conventional economic analysis (CEA)	0.28	0.93	−0.05	13.3
Extended economic analysis I (EEA I)	0.80	1.68	0.13	18.4
Extended economic analysis II (EEA II)	1.05	1.98	0.52	26.0
Socio-economic analysis (SEA)	1.15	2.17	0.58	27.8

incorporation of other external and indirect effects with no market prices (EEA II) such as the improvement of the quality of life and the groundwater recharge resulted in a further improvement with NPV of 1.05 MDT and IRR of 26.0%. However, the assessment of benefits related to the improvement of the quality of life is based on assumptions that still need to be verified through specific detailed surveys. As a matter of fact, the application of extended cost-benefit analysis (FORCES-MOD) is highly dependent on the quality of the available information on natural resources. The final step in the assessment is social cost-benefit analyses, which show slightly higher efficiency indicators (NPV of 1.15 MDT and IRR of 27.8%).

However, the level of profitability remains relatively low because of the heavy investments required in the field of soil and water conservation and the low agricultural production. Sensitivity tests on the discounting rate showed a certain stability of profitability criteria.

4. Conclusions

The water harvesting techniques are well adapted to their physical and social environment. They have various functions, e.g., water supplementation, flood prevention, water table recharge, and water erosion control. Because of these functions the economic profitability of their implementation and maintenance should be considered.

The importance of interdisciplinary and integrated approaches was revealed through the detailed impact assessment and economic evaluation of this study. In fact, the profitability of the WH works depends largely on the criteria chosen.

However, further refinements are needed to better include all possible impacts (positive and negative) that would occur as a result of the installation of the WH structures, e.g., in ecotourism, industry and tourism sector consumption. The interactions between upstream and downstream areas have to be addressed thoroughly to ensure equitable sharing of natural resources between different end users.

The use of geographical information systems and information technology would be of great value in transforming the results obtained into tools for decision-makers. The latter could be used to assess under which agro-ecological and socio-economic conditions investment in water harvesting measures could be a viable undertaking in dry Mediterranean areas or in other regions with similar agro-ecological conditions (southern Europe, Sahel).

Acknowledgements

This paper is a part of the INCO-DC project n°CT98-0269 '(WAHIA).' The financial support for travel from the Desert Research Foundation of Namibia (DRFN) and the United Nations University is highly appreciated.

References

Ben Mechlia, N. and Ouessar, M.: 2004, 'Water Harvesting Systems in Tunisia,' in: Th. Oweis and A. Bruggeman (eds), *Indigenous Water Harvesting Systems in West Asia and North Africa*, ICARDA Publications, Aleppo, Syria, pp. 21–41.

Biesemans, J.: 2000, 'Erosion Modelling as Support for Land Management in the Loss Belt of Flanders.' *Ph.D. Thesis*, Gent University, Belgium.

Derouiche, R.: 1997, *Contribution à l'étude par modèle numerique de l'impact des aménagements de CES sur la recharge de la nappe de Zeuss-Koutine.* Mémoire de in d'études, INAT, p. 68.

El Amami, Sl.: 1984, *Les aménagements hydrauliques traditionnels en Tunisie. CRGR*, Tunis.

Ennabli, N.: 1993, *Les aménagements hydrauliques et hydro-agricoles traditionnels en Tunisie*. Imprimerie Officielle de la République Tunisienne, Tunis, 255 pp.

FAO & World Bank: 1994, *Analyse économique de projets concernant les ressources naturelles en Tunisie. Une application du modèle FORCES-MOD aux bassins versants de Bou Hertma et de Marguellil*. Ministère de l'Agriculture, Tunis.

Fauck R., Makhlouf, Ez., Bachta, M.S., Laamary, M. and Marouani, A.: 1991, *Rapport d'évaluation sur les techniques de CES en Tunisie*. D-CES / Projet PNUD/FAO Tun 86-020.

Floret, C. and Pontanier, R.: 1982, *L'aridité en Tunisie présaharienne: Climat, sol, végétation et aménagement. ORSTOM*, Paris, p. 544.

Harris, H.: 1991, 'Implications of Climatic Variability,' in: H. Harris, P.J.M. Cooper, and M. Pala, (eds.), Proceedings of the Symposium *Soil and Crop Management for Improved Water Use Effeciency in Rainfed Areas*, 21–34, ICARDA, Syria, p. 357.

Hénia, L.: 1993, 'Climat et bilans de l'eau en Tunisie: Essais de régionalisation climatique par les bilans hydriques.' *Publication de l'Université de Tunis I, série Géographie,* p. 391.

Mini. Agri. (Ministère de l'Agriculture): 1990a, 'La stratégie nationale de la conservation des eaux et du sol (1990–2000). p. 29.

Mini. Agri. (Ministère de l'Agriculture): 1990b, *La stratégie nationale des ressources en eau (1990–2000).*

WAHIA 1999. *Opening Seminar report.* WAU, Wageningen.

WAHIA 2000. *First Annual Report.* WAU, Wageningen.